Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements is correct?

A. We can assume a "perfect sphere" model when analyzing the bending motion of a gymnast's wrist.
This is incorrect because detailed limb motion depends on shape and joint geometry, so a spherical approximation is not appropriate.

B. We can reasonably treat a parade float as a point mass when estimating how far it travels over several kilometers in a given time.
Over kilometer-scale motion, the object's size is negligible compared with its displacement, so a point-mass model is appropriate.

C. We are never allowed to model motion as if it occurred in a vacuum when we want to neglect air resistance.
This is incorrect because "vacuum" is a standard idealization used precisely to ignore drag effects when they are small.

D. None of the above is correct.
This is incorrect because option B describes a valid and commonly used approximation.

Q2. Which of the following equations is correct?

A. 1 dime = 1 $
This is incorrect because a dime is one tenth of a dollar, not one whole dollar.

B. 1 = 0.10
This is incorrect because it equates a dimensionless 1 to a different number without any units or conversion factor.

C. 1 dime = 0.10 $
This is correct because $1 = 10$ dimes, so one dime is $0.10\,\$.10 dimes = 0.10\$$
$This is incorrect because 10 dimes equals 1.00\,\$$, not $0.10\,\$$.

Q3. Which option has the incorrect unit?

A. Power: watt
This is correct because the SI unit of power is the watt (J/s).

B. Momentum: kg · m/s
This is correct because momentum has SI units of kg · m/s.

C. Electric potential: N/C
This is incorrect because N/C is the unit of electric field, while electric potential is J/C (volt).

D. Pressure: N/m^2
This is correct because pressure is force per area, N/m^2.

Q4. A rescue drone departs from the command post located at the origin and detects a signal beacon at the coordinates $\vec{r} = -5\,\hat{\imath} + 2\,\hat{\jmath}$, where length units are in hectometers. Which of the following statements is correct?

A. The distance between the beacon and the command post is 7 hectometers.
This is incorrect because the distance is found by the Pythagorean theorem, not by adding or rounding components arbitrarily.

B. The distance between the beacon and the command post is 5 hectometers.
This is incorrect because $\sqrt{(-5)^2 + 2^2} = \sqrt{29} \approx 5.4$, not 5.

C. The distance between the beacon and the command post is 2 hectometers.
This is incorrect because it confuses a single coordinate component with the full radial distance.

D. None of the above is correct.
This is correct because the distance is $\sqrt{29} \approx 5.4$ hectometers, which is not 7, 5, or 2.

Q5. If the displacement of an object is given by the function $\vec{s} = \left((0.50\,\mathrm{m/s^4})t^4 + (2.40\,\mathrm{m/s^2})t^2\right)\hat{\imath}$, which of the following statements is correct?

A. This is a constant-acceleration motion.
This is incorrect because the second derivative contains a t^2 term, so acceleration varies with time.

B. The magnitude of the acceleration is $2.40\,\mathrm{m/s^2}$.
This is incorrect because $2.40\,\mathrm{m/s^2}$ is a coefficient in $s(t)$ and not the (time-dependent) acceleration.

C. The magnitude of the acceleration at $t = 0$ is $0\,\mathrm{m/s^2}$.
This is incorrect because differentiating twice gives a nonzero constant term in $a(t)$.

D. The magnitude of the acceleration at $t = 0$ is $4.8\,\mathrm{m/s^2}$.
This is correct because $a(t) = \frac{d^2s}{dt^2} = (6.0t^2 + 4.8)\,\hat{\imath}$, so $|a(0)| = 4.8\,\mathrm{m/s^2}$.

Q6. An object takes 2.5 s to move from position A ($\vec{r}_A = 1.0\,\hat{\imath} - 3.0\,\hat{\jmath}$) to position B ($\vec{r}_B = -5.0\,\hat{\imath} + 4.0\,\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?

A. The moving distance is 9.2 m.
This is correct because the straight-line distance is $|\vec{r}_B - \vec{r}_A| = \sqrt{(-6)^2 + 7^2} = \sqrt{85} \approx 9.2\,\mathrm{m}$.

B. The speed at $t = 0$ must be $3.7\,\mathrm{m/s}$.
This is incorrect because knowing only endpoints and total time does not determine the instantaneous speed at $t = 0$.

C. The acceleration at $t = 0$ must be $1.5\,\mathrm{m/s^2}$.
This is incorrect because no information about how velocity changes with time is given, so acceleration cannot be inferred.

D. None of the above is correct.
This is incorrect because option A correctly computes the straight-line path length from the two position vectors.

Q7. A water droplet is launched from a fountain nozzle with speed v_0 at an angle θ above the horizontal. What is the speed of the droplet when it reaches its maximum height (ignore air resistance)?

A. v_0.
This is incorrect because the vertical component decreases to zero, so the total speed is less than v_0 (unless $\theta = 0$).

B. $v_0 \cos\theta$.
This is correct because at maximum height the vertical velocity is zero and the horizontal component remains $v_0 \cos\theta$.

C. $v_0 \sin\theta$.
This is incorrect because $v_0 \sin\theta$ is the initial vertical component, which becomes zero at the top.

D. 0.
This is incorrect because the droplet still has horizontal velocity at the top unless launched straight up.

Q8. A robotic cart is moving with an initial velocity $\vec{v}_0 = (12.0\,\text{m/s})\,\hat{\imath}$ when its motors engage, producing an acceleration $\vec{a} = (-0.80\,\text{m/s}^2)\,\hat{\imath} + (1.60\,\text{m/s}^2)\,\hat{\jmath}$. Based on this scenario, which of the following statements is correct?

A. The x-component of the cart's velocity remains constant.
This is incorrect because $a_x \neq 0$, so v_x changes with time.

B. The y-component of the cart's velocity remains constant.
This is incorrect because $a_y \neq 0$, so v_y changes with time.

C. Both the direction of motion and the speed change.
This is correct because the nonzero a_y changes direction and the nonzero a_x changes the speed (the velocity component along the original motion).

D. None of the above is correct.
This is incorrect because option C correctly describes the effect of having acceleration components in both axes.

Q9. If the magnitudes of both charges are quadrupled and their separation is doubled, what happens to the magnitude of the electrostatic force that each charge exerts on the other?

A. It stays the same.
This is incorrect because changing charges and distance changes $F \propto q_1 q_2 / r^2$ unless the factors exactly cancel.

B. It decreases by a factor of 4.
This is incorrect because the charges increase the force rather than decrease it.

C. It increases by a factor of 2.
This is incorrect because the combined scaling gives a larger factor than 2.

D. It increases by a factor of 4.
This is correct because $F \propto (4q)(4q)/(2r)^2 = 16/4 = 4$ times the original force.

E. It increases by a factor of 16.
This is incorrect because it accounts for the charge increase ($\times 16$) but ignores the distance-squared decrease ($\div 4$).

Q10. An object can escape a planet with radius R and mass M at an escape speed given by $v = \sqrt{\frac{2GM}{R}}$, where G is the universal gravitational constant. Based on this expression, which of the following statements is correct?

A. A more massive spaceship needs a larger escape speed from the same planet.
This is incorrect because the escape-speed formula depends on the planet's M and R, not the escaping object's mass.

B. A planet with the same mass but a larger radius requires a larger escape speed.
This is incorrect because increasing R makes $2GM/R$ smaller, which lowers the escape speed.

C. A planet with half the mass and half the radius always has a larger escape speed.
This is incorrect because halving both M and R leaves M/R unchanged, so the escape speed would be unchanged.

D. Escape speed is higher for a planet with larger mass and smaller radius.
This is correct because $v \propto \sqrt{M/R}$, so increasing M and decreasing R both increase v.

Q11. The force $\vec{F} = (5.20\,\text{N})\,\hat{\imath} - (1.60\,\text{N})\,\hat{\jmath}$ acts on a particle as it moves along the x axis from $x = -1.00\,\text{m}$ to $x = 2.50\,\text{m}$. What is the work, in joules, done by the force on the particle?

A. 19.0 J
This is incorrect because it comes from using $|\vec{F}|\Delta x$ instead of the dot product with the displacement direction.

B. 18.2 J
This is correct because only F_x contributes and $W = F_x \Delta x = 5.20 \times (2.50 - (-1.00)) = 5.20 \times 3.50 = 18.2\,\text{J}$.

C. 5.6 J
This is incorrect because it mistakenly uses the y-component magnitude even though there is no y displacement.

D. 0 J
This is incorrect because there is a nonzero force component along the direction of motion, so the dot product is not zero.

Q12. A 3.0 kg rubber ball initially moving at 4.0 m/s strikes a wall and rebounds at a speed of 2.5 m/s in the opposite direction. The collision lasts for 0.15 s. What is the average force exerted on the ball during the collision (magnitude)?

A. 130 N
This is correct because $F_{\text{avg}} = |\Delta p|/\Delta t = m|v_f - v_i|/\Delta t = 3.0 \times |(-2.5) - 4.0|/0.15 = 130\,\text{N}$.

B. 30 N
This is incorrect because it underestimates the impulse by not accounting for the velocity reversal.

C. 80 N
This is incorrect because it typically results from using the wrong collision time or the wrong change in velocity.

D. None of the above is correct.
This is incorrect because the impulse calculation gives a magnitude of 130 N, which is listed.

Q13. A 65 kg skateboarder is initially at rest. A 250 g ball moving at 30 m/s is caught and held (the ball sticks). What is the skateboarder's speed immediately after the catch?

A. 0.46 m/s
This is incorrect because it effectively ignores that the total mass after the catch is about 65 kg, which makes the final speed small.

B. 0.12 m/s
This is correct because momentum conservation for a perfectly inelastic catch gives $v = \frac{mv}{M+m} = \frac{0.250 \times 30}{65+0.250} \approx 0.115\,\text{m/s} \approx 0.12\,\text{m/s}$.

C. 115 m/s
This is incorrect because it comes from treating 250 g as 250 kg (a unit-conversion error).

D. 30 m/s
This is incorrect because it assumes the skateboarder ends up moving at the ball's original speed despite being far more massive.
